

MALLA REDDY UNIVERSITY

IV YEAR B.TECH – I SEMESTER

L/T/P/C

0/0/4/2

[MR22 – 1SKO1O2] CAREER ADVANCEMENT SKILLS

Common to all IV/I Branches

UNIT I : VERBAL & QUANTITATIVE SKILLS

1.By the time we reached the station, the train _____.

- A) left
- ~~B) had left~~
- C) leaves
- D) was leaving

2.Change the sentence into active voice:

“The work was completed by John.”

- A) John completes the work.
- ~~B) John completed the work.~~
- C) John is completing the work.
- D) John had completed the work.

3.Choose the synonym of “obstacle”:

- A) aid
- ~~B) barrier~~
- C) support
- D) opportunity

4.Arrange the following sentences to form a logical paragraph:

- 1.He decided to start his own business.
- 2.Rahul was unhappy with his job.
- 3.He worked there for five years.
- 4.He finally became successful.

- ~~A) 2, 3, 1, 4~~
- B) 3, 2, 1, 4
- C) 2, 1, 3, 4
- D) 1, 2, 3, 4

5. He has been working here _____ 2010.

- A) from
- B) for
- ~~C) since~~
- D) by

6. The average age of 3 students is 20 years. If one new student joins, the average becomes 22 years. What is the age of the new student?

- A) 24 years
- ~~B) 28 years~~
- C) 30 years
- D) 32 years

7. A number increases by 25% and then decreases by 20%. What is the net percentage change?

- ~~A) 0%~~
- B) 2% increase
- C) 5% decrease
- D) 10% increase

8. A can complete a work in 10 days and B in 15 days. How long will they take together to complete the work?

- A) 5 days
- ~~B) 6 days~~
- C) 8 days
- D) 9 days

9. A car travels 120 km at 60 km/h and returns at 40 km/h. What is the average speed?

- ~~A) 48 km/h~~
- B) 50 km/h
- C) 52 km/h
- D) 54 km/h

10. The average of five numbers is 30. If one number is 40, what is the average of the remaining four numbers?

- A) 25
- ~~B) 28~~
- C) 30
- D) 32

UNIT II : LOGICAL REASONING & ANALYTICAL SKILLS

11. Write the procedure to display the time angle between hour and minute hands for a given time (Clock problem).
12. Write a procedure to check if two strings are coded forms of each other (Coding-Decoding).
13. Write a procedure to solve a blood relation problem — given statements like “A is brother of B”.
14. Design a procedure that checks whether a given set of statements (premises) and a conclusion form a valid argument.
15. Write a procedure to find the day of the week for any given date (Calendar logic).

UNIT III : GIT & DEPLOYMENT

16. Initialize a new Git repository, add files, and commit them with proper messages.
17. Use Git commands to revert a commit and restore a previous version.
18. Automate deployment using a simple GitHub Action (CI/CD pipeline).
19. Resolve a merge conflict and document the steps you took.
20. Demonstrate Git workflow — create a new branch, switch between branches, and merge them.

UNIT IV : TEST – CASE DRIVEN PROGRAMMING AND CODE OPTIMIZATION

21. Why are hidden test cases important in automated evaluation systems?

- A) They help students memorize test data
- B) They check if code runs faster
- ☒ C) They verify the robustness of code against unseen inputs
- D) They are used only for syntax checking

22. Which of the following best describes input validation?

- A) Converting input to uppercase letters
- ☒ B) Ensuring the input data is correct and within expected limits
- C) Ignoring invalid data and continuing execution
- D) Handling runtime errors after they occur

23. What will happen if user input is not validated properly?

- A) The program will automatically correct it
- ☒ B) It may lead to runtime errors or security vulnerabilities
- C) It improves code performance
- D) The compiler will fix it during compilation

24. Which of the following is the correct use of try-except in Python?

- A) To define new variables
- ~~B) To handle exceptions and prevent abrupt program termination~~
- C) To test memory usage
- D) To create functions

25. Which principle helps make code modular and reusable?

- A) Writing all logic in one long function
- B) Using hard-coded values
- ~~C) Dividing the program into smaller functions and modules~~
- D) Ignoring comments and documentation

26. Which of the following practices improves code readability?

- ~~A) Using meaningful variable names~~
- B) Avoiding comments
- C) Writing everything in a single line
- D) Using random indentation

27. If a program's time complexity improves from $O(n^2)$ to $O(n \log n)$, it means:

- A) The program runs slower
- ~~B) The program runs faster for large inputs~~
- C) The program uses more memory
- D) The program is harder to understand

28. Which approach generally helps reduce memory usage?

- A) Storing duplicate data
- B) Using more temporary variables
- ~~C) Reusing variables and data structures efficiently~~
- D) Avoiding loops

29. Which of the following combines all good programming practices?

- A) Code that passes only sample test cases
- ~~B) Code that is fast, readable, modular, and handles all inputs safely~~
- C) Code that uses complex logic even for simple tasks
- D) Code that ignores hidden cases for speed

30. Which of the following checks best represents input validation?

- ~~A) if input_value < 0: print("Invalid input")~~
- B) input_value = int(input())
- C) print(input_value)
- D) input_value += 1

UNIT V : VERSANT & CORPORATE ENGLISH

31. Explicit email etiquette. Discuss why it is essential in today's workplace communication.
32. Illustrate group discussion in detail?
33. Explain why do companies conduct group discussions during recruitment?
34. Define JAM. Explain the main objective of a JAM session during interview?
35. Discuss the purpose of subject line in the email.